

Agent Tool Surface & Return Signatures

This document inventories every programmatic tool exposed to the Macro Propose, Micro Propose, and Re-Crop Agents across both passes of the region discovery and quality verification lifecycle.

Each tool is detailed with its operational purpose, its calling schema, and the exact JSON responses/error messages it returns to the agent's multimodal chat context.

register_census

Registers the agent's structured Page Census checklist on-disk inside the private page partition file to anchor cognitive planning.

Purpose

- Exposed in both Pass 1 and Pass 2 to serialize planned targets to disk.
- Enforces a strict chronological planning step on Turn 1 before any visual exploration or drafting can begin.

Verbatim API Description

Sent to the model via the `tools` parameter on every API call:

```
Register your structured Page Census checklist on-disk.  
– Use this tool EXACTLY ONCE on your second turn (Turn 1: Attention &  
Census Mapping) to serialize your planned targets to disk.  
– You will be programmatically blocked from exiting the page until all  
registered items are COMPLETED via draft_region.
```

Return Signatures

Success Response (Pass 1 & Pass 2)

```
{  
  "ok": true,  
  "status": "census_registered",  
  "message": "Census checklist has been persisted to disk. You must now  
zoom and draft every item sequentially. You will be programmatically
```

```
blocked from exiting the page until all items are COMPLETED."
}
```

Failure Response

```
{
  "ok": false,
  "error": "CensusValidationError: items list is malformed, missing
sequential IDs, or duplicate semantic labels."
}
```

get_grid_overview

Determines spatial orientation by rendering the full PDF page (Pass 1) or the local parent Macro-Object view (Pass 2) with a coordinate grid overlay.

Purpose

- Exposed in both Pass 1 and Pass 2 to establish initial canvas boundaries.
- Enforces alphanumeric coordinate grid cell labels (A1–J10) and axis markers (0.0 to 1.0).

Verbatim API Description

Sent to the model via the `tools` parameter on every API call:

```
Render a full page or a parent macro drawing canvas with a normalized
coordinate grid overlaid.
- Use this tool EXACTLY ONCE at the very beginning of the session to
locate boundaries.
- Axis labels 0.0–1.0; cell labels A1, A2, ... starting top-left.
```

Return Signatures

Success Response (Pass 1 & Pass 2)

Returns a text part carrying JSON metadata side-by-side with the rendered grid JPEG image payload.

```
{
  "ok": true,
  "saved_path": "<project_root>/crops/_grid/3048_CAD_159825_ZUSAMMENBAU_3048_p02_grid_overlay.png"
}
```

Failure Response

```
{
  "ok": false,
  "error": "PageRangeError: requested page 159 exceeds document page count of 158."
}
```

zoom_into_region

Acts as a high-resolution microscopic magnifying glass to inspect dense drawing details, trace leader lines, or check margin boundaries.

Purpose

- Purely exploratory; **DOES NOT** commit or save the region to the verification queue.
- Allows the agent to visual-audit corners and line ends before deciding whether to call `draft_region`.

Verbatim API Description

Sent to the model via the `tools` parameter on every API call:

```
Render a high-resolution exploratory preview of a specific coordinate box [x1, y1, x2, y2].
- Use this tool strictly during Microscope Investigation to inspect the boundaries of your primary target object.
- WARNING: You MUST remain tightly anchored to the main target itself. Do NOT let your attention drift to trace or follow other adjacent objects, notes, or border texts. This is purely exploratory and DOES NOT save or commit a region.
```

Return Signatures

Success Response (Pass 1 & Pass 2)

Returns a text part carrying a logically balanced, action-forcing JSON instruction side-by-side with the high-resolution JPEG image payload. This completely de-triggers faking by avoiding any `"ok": true` response strings.

```
{
  "status": "investigation_performed",
  "message": "This is a temporary high-resolution preview and is NOT
saved or committed. You must now make a decision: 1) IF this zoomed
view is correct and cleanly framed, you MUST call the 'draft_region'
tool with these coordinates to formally commit and register this
region. 2) IF this zoomed view is incorrect or unsatisfactory, you MUST
adjust your coordinates. You can call 'zoom_into_region' again to re-
inspect. Do not mark as [COMPLETED] until 'draft_region' is
successfully executed."
}
```

Failure Response

```
{
  "status": "investigation_failed",
  "error": "BadBoxError: Coordinate box is invalid (wrong shape,
degenerate, or inverted: got [0.82, 0.5, 0.82, 0.9])."
}
```

draft_region

Formally registers and commits a candidate coordinate box into the programmatic verification and re-cropping queue.

Purpose

- Saves the high-resolution crop to the temporary `_scratch/` folder.
- Creates a strongly typed `CropDraft` object and registers it in the RAM-based drafts checklist. This is the **only** tool that moves a crop downstream to Pass 2 Verification.

Verbatim API Description

Sent to the model via the `tools` parameter on every API call:

```
Render and formally commit a candidate coordinate box to the
verification queue.
- Use this tool strictly during Committal to permanently register a
fully verified region.
- This is the ONLY tool that successfully saves a crop for verification
and cataloging.
```

Return Signatures

Success Response

```
{
  "ok": true
}
```

Failure Response

If coordinates are degenerate, inverted, or exceed pixel cap parameters. Note that if the crop exceeds the strict 40-Megapixel limit on oversized sheets, the rendering engine dynamically auto-scales the DPI instead of failing.

```
{
  "ok": false,
  "error": "BadBoxError: Coordinate box is inverted or out of bounds."
}
```

skip_page

Declares that the current page carries absolutely zero technical value and should be completely bypassed.

Purpose

- Only exposed to the Pass 1 Macro Propose Agent.

- Allows the agent to cleanly bypass cover sheets, prose narrative logs, or purely descriptive introductions without wasting rounds of verification.

Verbatim API Description

Sent to the model via the `tools` parameter on every API call:

```
Declare that this page contains absolutely zero technical content,
mechanical drawings, or tables, and must be skipped.
- Use this tool strictly during Committal as a rare exception, only
when the CROPPING RULES explicitly dictate skipping.
```

Return Signatures

Success Response

```
{
  "ok": true,
  "skipped": true
}
```

Failure Response

```
{
  "ok": false,
  "error": "SkipPageError: Skip page called on already drafted canvas."
}
```

skip_macro_view

Declares that the current Macro-Object view contains absolutely zero descriptive text notes with leader lines or arrows, and should be completely bypassed.

Purpose

- Only exposed to the Pass 2 Micro Propose Agent.
- Allows the agent to cleanly bypass pure-visual drawings with zero descriptive annotations without triggering exit-enforcement loops.

Verbatim API Description

Sent to the model via the `tools` parameter on every API call:

Declare that this Macro-Object view contains absolutely zero descriptive text notes with leader lines or arrows, and must be skipped.

- Use this tool strictly during Committal as a rare exception, only when the view is purely visual with zero annotations.

Return Signatures

Success Response

```
{
  "ok": true,
  "skipped": true
}
```

Failure Response

```
{
  "ok": false,
  "error": "SkipPageError: Skip macro view called on already subdivided canvas."
}
```